

Classification Learner

The screenshot shows the 'New Session from Workspace' dialog in the MATLAB Classification Learner app. The interface is divided into several sections:

- Data set:** Contains a 'Data Set Variable' dropdown set to 'meas' (150x4 double). Below it are two radio buttons: 'Use columns as variables' (selected) and 'Use rows as variables'. The 'Response' section has two radio buttons: 'From data set variable' and 'From workspace' (selected). Below this is a dropdown for 'species' (150x1 cell, 3 ...).
- Predictors:** A table with columns 'Name', 'Type', and 'Range'. It lists four predictors: 'column_1', 'column_2', 'column_3', and 'column_4', all of type 'double'. Below the table are 'Add All' and 'Remove All' buttons, and a 'Refresh' button.
- Validation:** Contains a 'Validation Scheme' dropdown set to 'Cross-Validation'. Below it is a text description: 'Protects against overfitting. For data not set aside for testing, the app partitions the data into folds and estimates the accuracy on each fold.' There is a 'Cross-validation folds' spinner set to 5, and a link 'Read about validation'.
- Test:** Contains a checkbox 'Set aside a test data set' which is unchecked. Below it is a 'Percent set aside' spinner set to 10. A text description explains that a test set is used to evaluate model performance after tuning. There is a link 'Read about test data'.

At the bottom right are 'Start Session' and 'Cancel' buttons. At the bottom left is a link 'How to prepare data'.

	Name	Type	Range
☒	column_1	double	4.3 .. 7.9
☒	column_2	double	2 .. 4.4
☒	column_3	double	1 .. 6.9
☒	column_4	double	0.1 .. 2.5

You are telling MATLAB:

“Use these numbers (features) to predict this label (class).”

Data Set Variable: meas (150×4 double)

- 150 samples (flowers)
- 4 numbers per sample (features)
- Each row = one flower
- Each column = one measurement (length/widths)

✓ **Keep: “Use columns as variables”**

Means: each column is a feature → correct.

Response (this is the MOST important part)

From workspace → species (150×1 cell)

- This is **what you want to predict**

- 150 labels, one per flower
- Values are strings:

setosa, versicolor, virginica

👉 This turns the problem into **classification**

(not regression).

If response were numbers → regression

If response were categories →
classification

You're doing classification ✓

Predictors (inputs to the model)

You see:

- column_1
- column_2
- column_3
- column_4

Each is:

- type: double
- range: min–max values

These are the **inputs X**

species is the **output y**

Right side:

Validation

Cross-Validation (5 folds)

Means:

- Split data into 5 parts
- Train on 4, test on 1
- Rotate and average accuracy

This is **standard**, don't touch it now.

Test section

Unchecked → fine.

Means you're not keeping a separate test set.

Cross-validation already handles evaluation.

What happens when you click

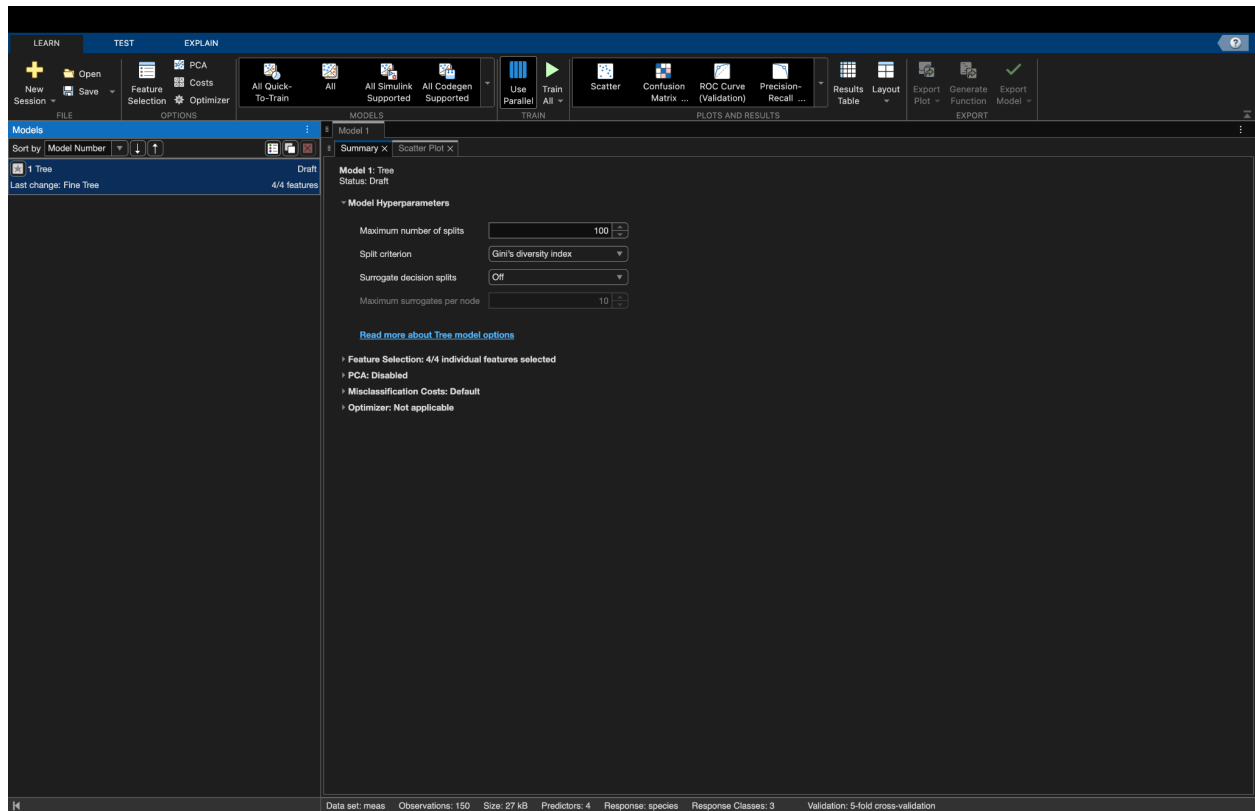
Start Session

MATLAB will:

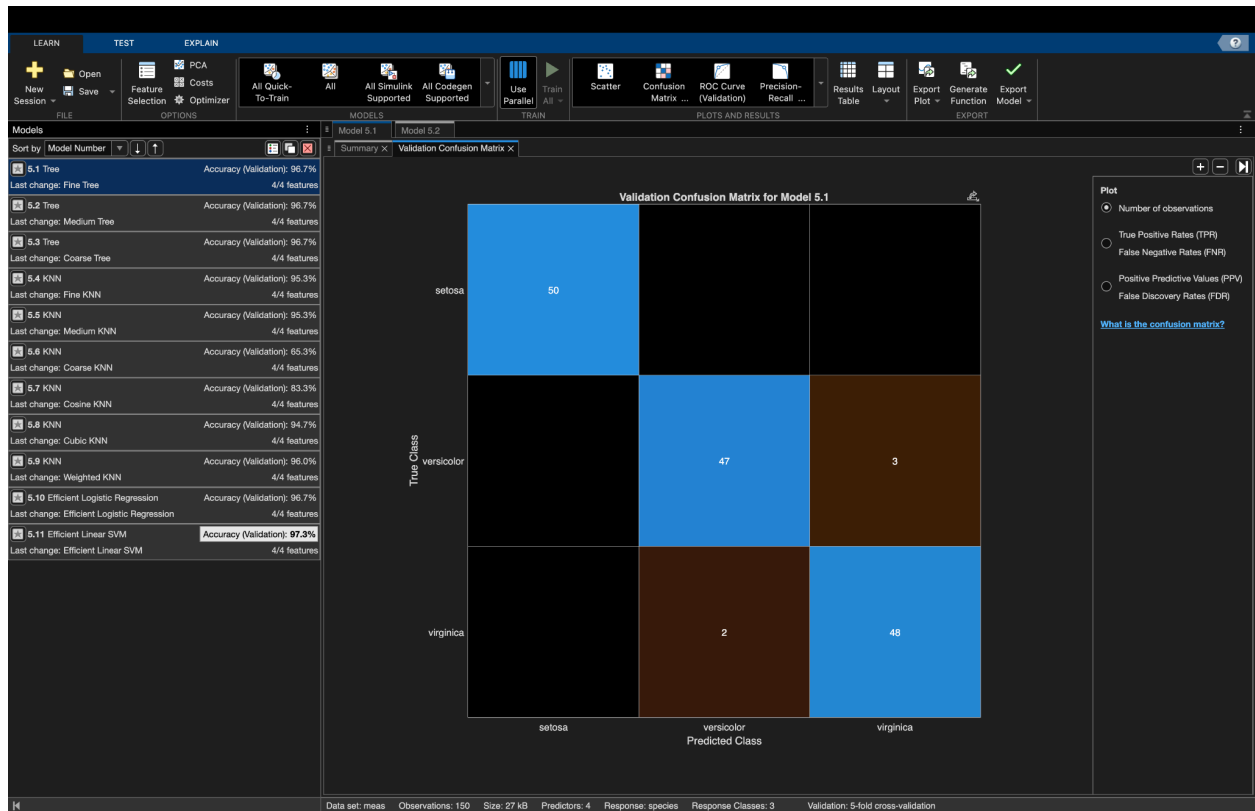
1. Recognize this as **classification**
2. Let you try **multiple classifiers**
3. Compare accuracy
4. Let you tune hyperparameters
5. Let you **export MATLAB code**



After starting the new session
You will get this graph, all the dots are the
samples. Different colours indicate
different classes.
Matlab gives a tree model in start
itself(auto)



These are the hyperparameters u can tune bro



I deleted that tree model and clicked
All-Quick-To-Train

This takes all simple models and trains
them on the dataset and provides all
inferences

This is osm!

Left panel (Models list)

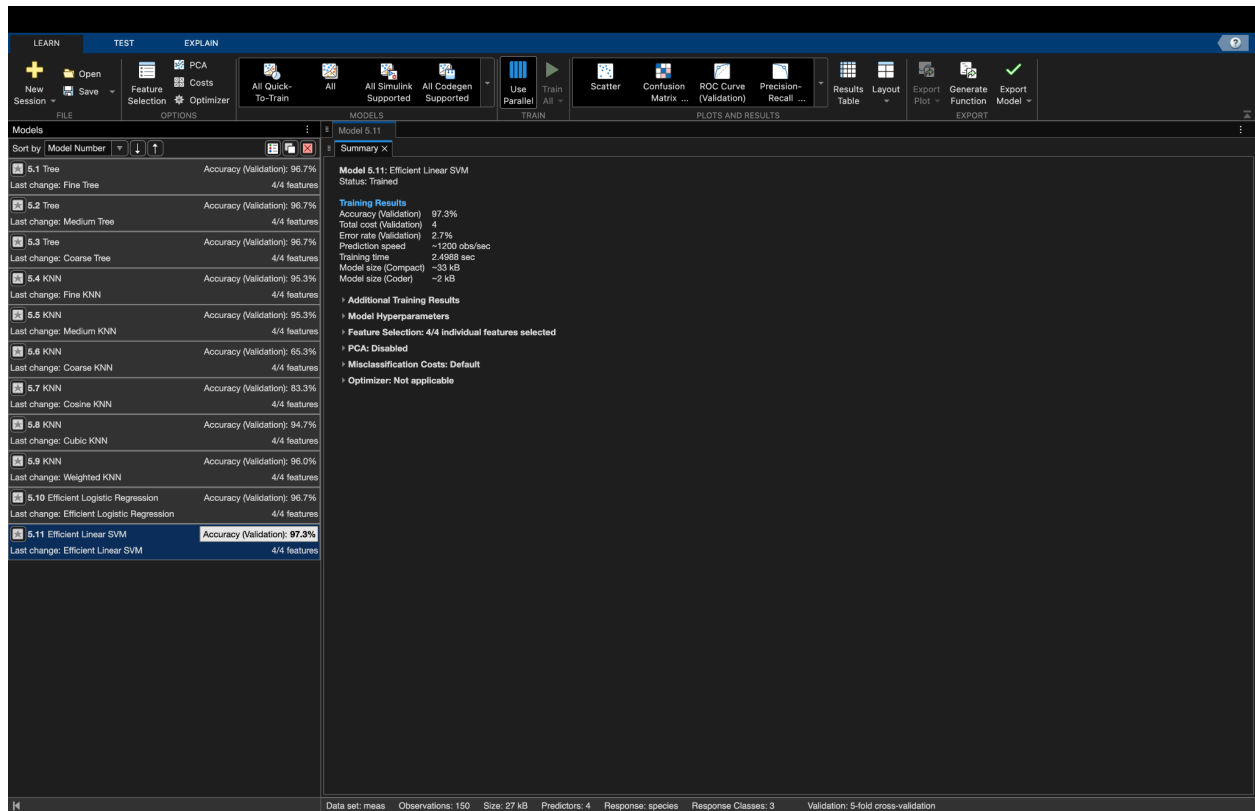
**MATLAB auto-trained many models for
you:**

- Decision Trees (fine / medium / coarse)
- kNN (different distance metrics)
- Logistic Regression
- Linear SVM

Each line shows:

- **Validation accuracy** (cross-validated, not cheating)
- Same dataset, same splits → fair comparison

This is **model selection**, not magic.



Linear SVM beats decision trees

This is because of the data geometry

Trees can overfit slightly and are sensitive to small variations

“Decision trees are powerful, so they should win”

Not always.

Power $\neq$ best for every data distribution

Data Type	Best Model
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Linear / smooth	SVM / Logistic Regression
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Axis-align ed rules	Trees
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Complex interactio ns	Random Forest / XGBoost
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High noise	Ensembles
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“SVM outperformed trees because the Iris classes are nearly linearly separable, making margin-based classifiers more suitable than axis-aligned splits.”

The screenshot displays the Orange3 machine learning interface. The 'Models' list on the left shows various models, with '5.11 Efficient Linear SVM' highlighted as the top performer with an accuracy of 97.3%. The right pane provides detailed information for this model, including training results and hyperparameters.

Model	Accuracy (Validation)	Features
5.1 Tree	96.7%	4/4 features
5.2 Tree	96.7%	4/4 features
5.3 Tree	96.7%	4/4 features
5.4 KNN	95.3%	4/4 features
5.5 KNN	95.3%	4/4 features
5.6 KNN	65.3%	4/4 features
5.7 KNN	83.3%	4/4 features
5.8 KNN	94.7%	4/4 features
5.9 KNN	96.0%	4/4 features
5.10 Efficient Logistic Regression	96.7%	4/4 features
5.11 Efficient Linear SVM	97.3%	4/4 features

Model 5.11: Efficient Linear SVM
Status: Trained

Training Results

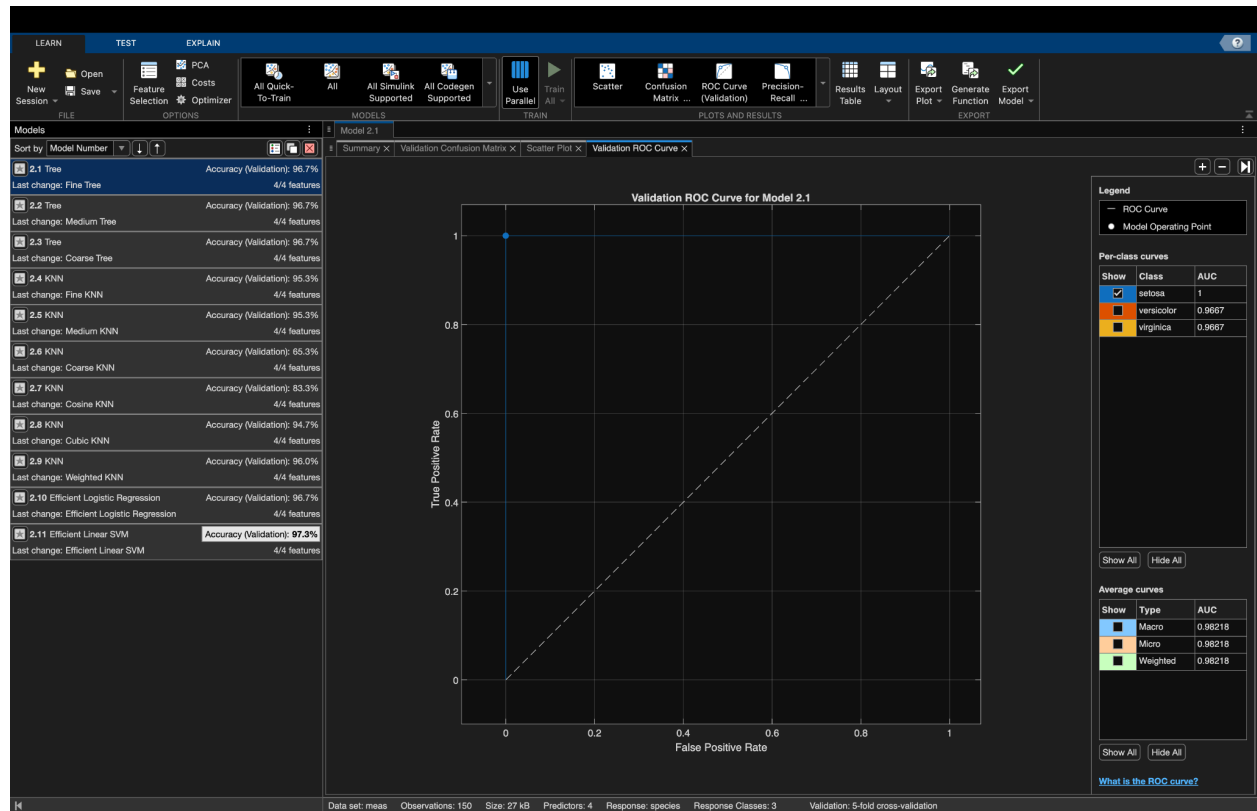
Accuracy (Validation)	97.3%
Total cost (Validation)	4
Error rate (Validation)	2.7%
Prediction speed	~1000 obs/sec
Training time	2.4988 sec
Model size (Compact)	~33 kB
Model size (Coder)	~2 kB

Additional Training Results

- Model Hyperparameters
 - Preset: Efficient Linear SVM
 - Learner: SVM
 - Solver: Auto
 - Regularization: Auto
 - Regularization strength (Lambda): Auto
 - Relative coefficient tolerance (Beta tolerance): 0.0001
 - Multiclass coding: One-vs-One
- Feature Selection: 4/4 individual features selected
- PCA: Disabled
- Misclassification Costs: Default
- Optimizer: Not applicable

Data set: meas Observations: 150 Size: 27 kB Predictors: 4 Response: species Response Classes: 3 Validation: 5-fold cross-validation

These are the hyperparameters u can change



U can view roc curves